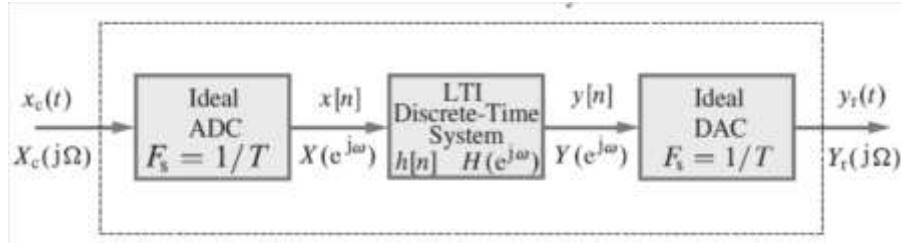


- Q1** An analog signal $x_c(t) = 5 \sin(2\pi 250t) + 10 \sin(2\pi 300t)$ is to be processed using the system shown in figure 1.1 in which the sampling frequency is 1 kHz. Design a minimum-order IIR digital filter that will suppress the 300 Hz component down to 20 dB while pass the 250 Hz component with attenuation of 2 dB. Determine the system function of the filter and plot its log-magnitude response in dB. Draw the efficient filter realization structure. Justify. **15**



- Q2** A system for the discrete-time spectral analysis of continuous-time signal is shown in figure 2.1. **10**

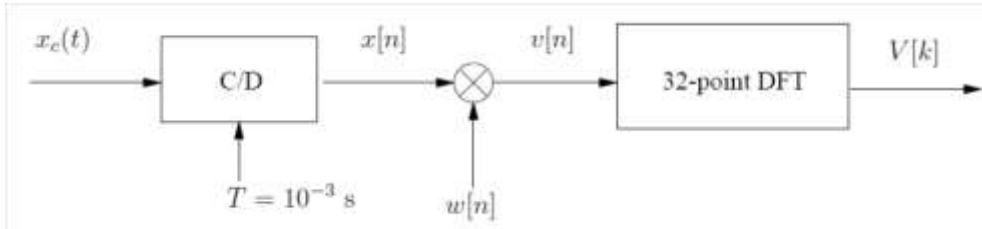


Figure 2.1: Spectral analysis system

$w[n]$ is rectangular window of length 32. $w[n] = \begin{cases} \frac{1}{32}, & 0 \leq n \leq 31 \\ 0 & \text{otherwise} \end{cases}$

Listed below are ten signals, at least one of which was the input $x_c(t)$. Indicate which signal(s) could have been the input $x_c(t)$ which produced the plot of $|V(k)|$ shown in dB units in Figure 2.2. As always, provide reasoning for your choice(s). Choice of inputs:

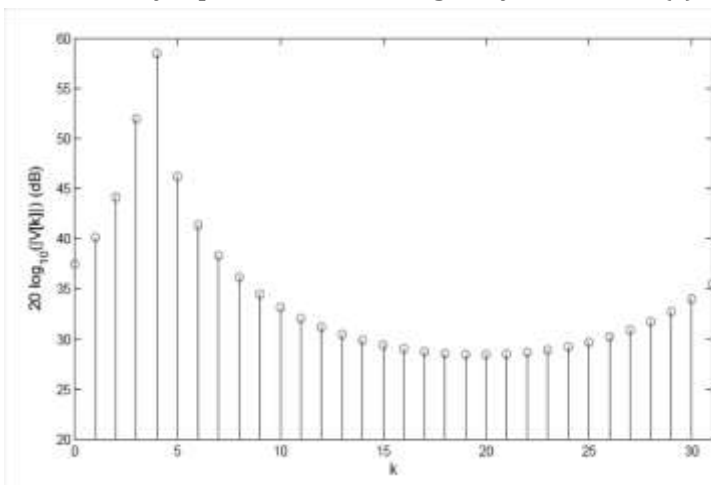


Figure 4.2: Output $|V(k)|$ in dB

1. $x_1(t) = 1000 \cos(230\pi t)$
2. $x_2(t) = 1000 \cos(115\pi t)$
3. $x_3(t) = 10e^{j460\pi t}$
4. $x_4(t) = 1000e^{j230\pi t}$
5. $x_5(t) = 10e^{j230\pi t}$
6. $x_6(t) = 1000e^{j250\pi t}$
7. $x_7(t) = 10 \cos(250\pi t)$
8. $x_8(t) = 1000 \cos(218.75\pi t)$
9. $x_9(t) = 10e^{j200\pi t}$
10. $x_{10}(t) = 1000 \cos(230\pi t)$

- Q3** The system in figure 3.1 computes an N-point (where N is an even number) DFT $X[k]$ of an N-point sequence $x[n]$ by decomposing $x[n]$ into two $N/2$ -point sequences $g_1[n]$ and $g_2[n]$, computing the $N/2$ -point DFT's $G_1[k]$ and $G_2[k]$, and then combining these to form $X[k]$. If $g_1[n]$ is the even-indexed values of $x[n]$ and $g_2[n]$ is the odd-indexed values of $x[n]$ i.e. $g_1[n] = x[2n]$ and $g_2[n] = x[2n + 1]$ then $X[k]$ will be the DFT of $x[n]$. **10**

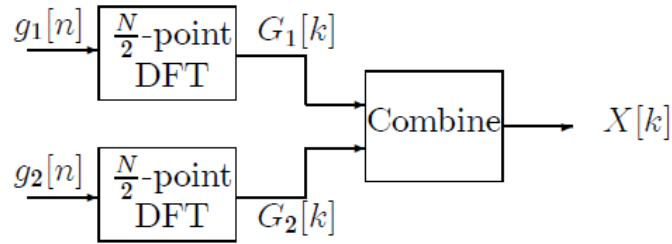


Figure 3.1

In using the system in figure 3.1 an error is made in forming $g_1[n]$ and $g_2[n]$, such that $g_1[n]$ is incorrectly chosen as the odd-indexed values and $g_2[n]$ as the even indexed values but $G_1[k]$ and $G_2[k]$ are still combined as in figure 3.1 and the incorrect sequence $\hat{X}[k]$ results. Express $\hat{X}[k]$ in terms of $X[k]$

- Q4** Design a FIR linear-phase, digital filter approximating the ideal frequency response using windowing technique. Plot the magnitude and phase response of filter and advice accordingly. **15**

$$H_d(e^{j\omega}) = \begin{cases} 1, & \text{for } |\omega| \leq \frac{\pi}{6} \\ 0, & \text{for } \frac{\pi}{6} \leq |\omega| \leq \frac{\pi}{3} \\ 1, & \text{for } \frac{\pi}{3} \leq |\omega| \leq \pi \end{cases}$$

Note: Use minimum 3 (diverse) windows, and 25-tap filter for comparison.

- Q5** Design a FIR filter that completely blocks the frequency $\omega_c = \frac{\pi}{4}$, by suitable placing poles and zeros in the z-plane. The constructed filter should yield real output given that the input is real. **10**
- Draw poles and zeros of the filter in the z-plane. Clearly state the magnitude and angle.
 - Calculate the system function $H(z)$ and frequency response $H(e^{j\omega})$
 - Sketch the magnitude response
 - If input signal to the filter is given by $x[n] = 10 + 3 \cos\left(\frac{\pi}{4}n\right) + \sin\left(\frac{\pi}{3}n + \frac{\pi}{2}\right)$, $-\infty < n < \infty$. Determine the output $y[n]$ of the filter.